

Midterm Exam Subjects : Computer Architecture (Close Book)

1. Please refer to the development process of Intel microprocessors, and try to speculate that with the advancement of integrated circuit process technology, processors consume less and less power, but why do most processors need to use cooling fans?
2. What is a command set computer? How does such a concept help or benefit computer design?
3. Please explain the following terms: (1) Response time (2) Execution time (3) Throughput (4) Benchmark
4. The performance of a P microprocessor with 100 MHz was evaluated by executing 10,000,000 evaluation program instructions, and it was found that it took 0.25 seconds. What are the CPI and MIPS data during this performance experiment?
5. There is a 500MHz processor, and the following is the data obtained after measurement

Instruction set	CPI	frequency of use
A	2	40%
B	3	25%
C	3	25%
D	3	10%

- (1) What is the MIPS of this processor, please?
 - (2) What is the CPI of this processor, please?
6. Please try to explain the program compilation period (compile time)What are the main tasks included? Why does a program have to go through this stage before it can be executed?
 7. Using the stacked structure, the diagram illustrates the process of generating post-fix representations in the following formulas (a) and pre-fix representations in the following formulas (b)

(a) $A / B - C + D * E + F * G$

(b) $A * (B + C) - D$